SMART@Clinic | Hackathon Appendix

Supporting technical reference for the 3-minute prototype demonstration

A. Specialty-specific dashboard architecture

- 20 specialty families and 113 clinical-focus configurations are documented.
- Each configuration identifies priority metrics, supporting information, and recommended visualizations.
- Provider specialty determines initial defaults; physician preferences and patient-level settings refine the display.
- The architecture is designed for future support of multiple conditions per patient.
- Implementation status: the interactive prototype demonstrates asthma, migraine, and diabetes pathways; the broader catalog is an architecture reference.

B. Visualization documentation

Clinical focus	Key measurements	Recommended visuals	Status
Asthma	Symptom days; rescue medication; FEV1; medication reports	Symptom-frequency bars; longitudinal FEV1; medication reporting summary	Prototype pathway
Headache & migraine	Headache days; severity; acute medication; activity limitation	Weekly headache-burden trend; severity trend; sleep-versus-severity scatterplot	Prototype pathway
Diabetes	Spot glucose; reported low-glucose events; medication reports	Glucose trend; daily reading distribution; event-frequency bars	Prototype pathway
Heart failure	Weight; symptoms; blood pressure; functional status	Weight trend; symptom timeline; blood-pressure chart	Proposed
Epilepsy	Seizure frequency; duration; medication reports	Event timeline; weekly seizure counts	Proposed
Inflammatory bowel disease	GI symptoms; flare reports; functional impact	Symptom-burden trend; flare timeline	Proposed
Behavioral health	Mood; sleep; functioning; symptom scores	Longitudinal score trends; sleep-versus-mood scatterplot	Proposed

The table shows representative examples, not all 113 configurations. Proposed visualizations are not claimed as implemented.

C. Predictive modeling & historical pattern discovery

Symptom trajectory forecasting

- Explores short-term symptom burden using historical patient-reported measurements.
- Aggregates to interpretable weekly measures where appropriate, rather than projecting isolated daily spikes.
- Distinguishes observed history from exploratory projection and uncertainty.
- Withholds projections when data are insufficient or the model does not provide useful performance.
- Does not predict diagnoses, emergencies, or treatment outcomes.

Historical pattern discovery

- Examines paired observations such as sleep duration versus headache severity.
- Reports sample size and descriptive relationships; patterns are not causal evidence.
- Missing reports remain missing; unreported symptoms are not assumed absent.
- Calculations use synthetic demonstration data and are not clinically validated.

D. Chatbot, dictation & billing workflow

1. Patient data and trends: Patient reports, device measurements, and longitudinal visualizations.

2. Clinical chatbot: Proposed conversational questions grounded in authorized patient records, with source references.

3. Physician dictation and review: Physician captures, checks, and approves documentation.

4. Billing assistance: Proposed documentation-supported billing code suggestions for physician review; no automatic submission.

Safety boundary: chatbot answers require source grounding and authorized access. Billing suggestions require coding-rule checks, documentation support, and human approval.

E. Technical feasibility and limitations

Component	Approach / status
Frontend	HTML, CSS, JavaScript
Personalization	Specialty defaults; provider and patient overrides
Charts	Interactive SVG visualizations
Analytics	Local statistical calculations on synthetic records
Documentation	Browser text entry and optional dictation
Chatbot	Proposed patient-data-grounded conversational interface
Billing	Proposed clinician-reviewed coding assistant
Clinical records	Fictional data; no live EHR connection

Production prerequisites: secure authentication, role-based authorization, privacy controls, approved EHR integration, validation of clinical analytics, and compliant coding workflows.

Full technical reference

Companion reference: SMART@Home Complete Specialty Metrics Directory — full list of 20 specialty families and 113 clinical-focus presets, with key metrics and supporting information.